

BUI-E1: Buildmanagement

1. Buildmanagement mit Maven

Warum habe ich dieses Werkzeug gewählt?

Maven wurde gewählt, weil es ein bekanntes Buildmanagement-System für Java-Projekte ist. Es kann den Buildprozess automatisieren und über eine zentrale Datei `pom.xml` gesteuert werden.

Welche Befehle habe ich ausgeführt?

- `mvn -version`
- `mvn compile`
- `java -cp target/classes Main`
- `java -cp target/classes TaskManagerManualTest`

```

C:\Users\zaid>java -version
openjdk version "21.0.5" 2024-10-15 LTS
OpenJDK Runtime Environment Temurin-21.0.5+11 (build 21.0.5+11-LTS)
OpenJDK 64-Bit Server VM Temurin-21.0.5+11 (build 21.0.5+11-LTS, mixed mode, sharing)

C:\Users\zaid>javac -version
javac 21.0.5

C:\Users\zaid>cd "C:\Users\zaid\Documents\Smart Task Manger"

C:\Users\zaid\Documents\Smart Task Manger>dir
Volume in drive C has no label.
Volume Serial Number is E013-E4AD

Directory of C:\Users\zaid\Documents\Smart Task Manger

06/14/2026  04:59 PM    <DIR>          .
06/13/2026  12:07 AM    <DIR>          ..
06/14/2026  01:02 AM                1,584 README.md
06/14/2026  12:59 AM    <DIR>          src
                1 File(s)                1,584 bytes
                3 Dir(s)  49,257,283,584 bytes free

C:\Users\zaid\Documents\Smart Task Manger>dir src
Volume in drive C has no label.
Volume Serial Number is E013-E4AD

Directory of C:\Users\zaid\Documents\Smart Task Manger\src

06/14/2026  12:59 AM    <DIR>          .
06/14/2026  04:59 PM    <DIR>          ..
06/14/2026  04:59 PM                6,115 ConsoleUI.java
06/14/2026  12:49 AM                1,183 InputValidator.java
06/14/2026  12:31 AM                 211 Main.java
06/14/2026  12:31 AM                 55 Priority.java
06/14/2026  04:59 PM                1,020 Task.java
06/14/2026  12:31 AM                1,883 TaskManager.java
06/14/2026  12:59 AM                3,884 TaskManagerManualTest.java
06/14/2026  12:31 AM                 80 TaskStatus.java
                8 File(s)                14,431 bytes
                2 Dir(s)  49,217,658,880 bytes free

C:\Users\zaid\Documents\Smart Task Manger>|

```

Abbildung 1: Java-Version und Dateien des Smart-Task-Manager-Projekts im Terminal

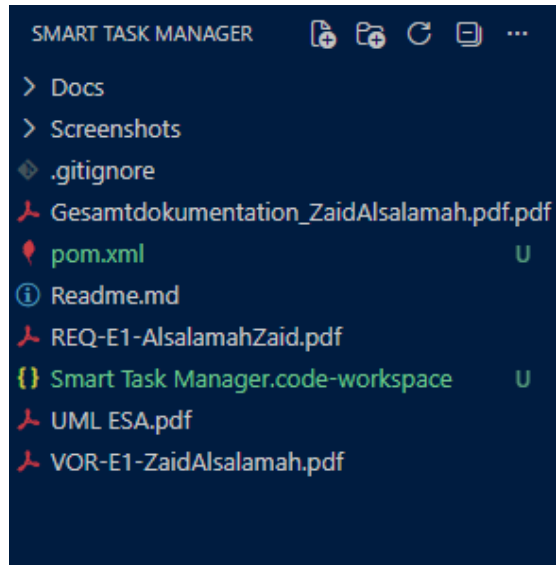


Abbildung 2 : erstellen wir eine Maven-Datei

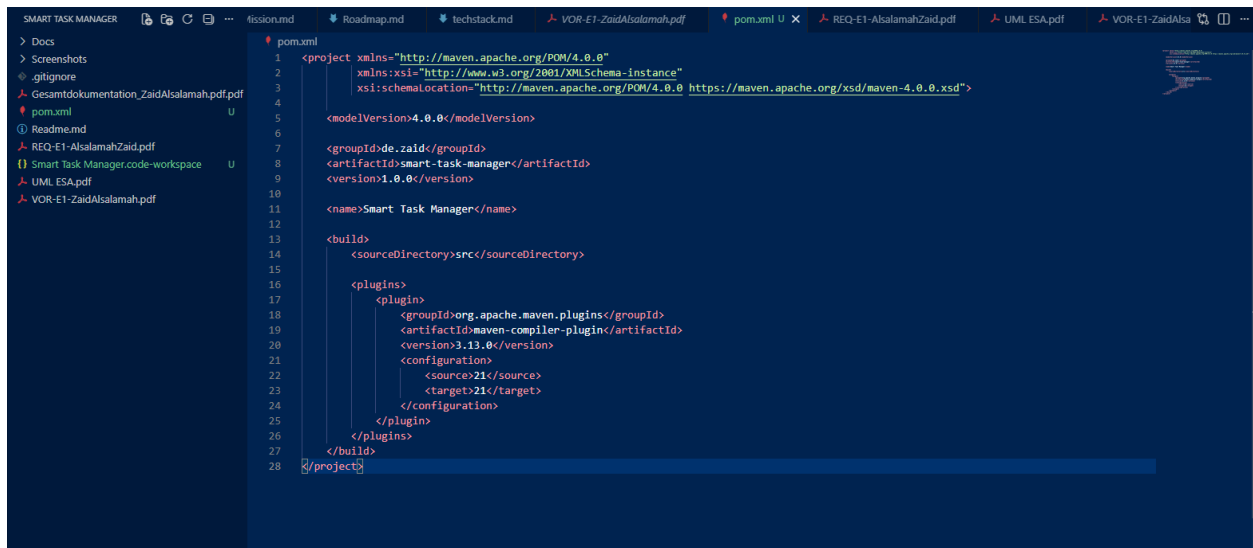


Abbildung 3 : Inhalt Maven-Datei

```

PS C:\Users\zaid\Desktop\Smart Task Manager> $env:MAVEN_HOME="C:\Tools\apache-maven-3.9.16"
PS C:\Users\zaid\Desktop\Smart Task Manager> $env:PATH="$env:MAVEN_HOME\bin;$env:PATH"
PS C:\Users\zaid\Desktop\Smart Task Manager> mvn -version
Apache Maven 3.9.16 (2bdd9fdd4b155e6f800e807eb73fd829a51d5)
Maven home: C:\Tools\apache-maven-3.9.16
Java version: 21.0.5, vendor: Eclipse Adoptium, runtime: C:\Program Files\Eclipse Adoptium\jdk-21.0.5.11-hotspot
Default locale: en_US, platform encoding: UTF-8
OS name: "windows 11", version: "10.0", arch: "amd64", family: "windows"
PS C:\Users\zaid\Desktop\Smart Task Manager> mvn compile
[INFO] Scanning for projects...
[INFO]
[INFO] -----< de.zaid:smart-task-manager >-----
[INFO] Building Smart Task Manager 1.0.0
[INFO] from pom.xml
[INFO] -----[ jar ]-----
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-resources-plugin/3.4.0/maven-resources-plugin-3.4.0.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-resources-plugin/3.4.0/maven-resources-plugin-3.4.0.pom (7.9 kB at 23 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-resources-plugin/3.4.0/maven-resources-plugin-3.4.0.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-resources-plugin/3.4.0/maven-resources-plugin-3.4.0.jar (31 kB at 1.1 MB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-compiler-plugin/3.13.0/maven-compiler-plugin-3.13.0.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-compiler-plugin/3.13.0/maven-compiler-plugin-3.13.0.pom (10 kB at 402 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-compiler-plugin/3.13.0/maven-compiler-plugin-3.13.0.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-compiler-plugin/3.13.0/maven-compiler-plugin-3.13.0.jar (83 kB at 2.5 MB/s)
[INFO] --- resources:3.4.0:resources (default-resources) @ smart-task-manager ---
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/shared/maven-filtering/3.4.0/maven-filtering-3.4.0.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/shared/maven-filtering/3.4.0/maven-filtering-3.4.0.pom (6.5 kB at 249 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/shared/maven-shared-components/43/maven-shared-components-43.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/shared/maven-shared-components/43/maven-shared-components-43.pom (3.8 kB at 135 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/shared/maven-shared-components/43/maven-shared-components-43.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/shared/maven-shared-components/43/maven-shared-components-43.jar (3.8 kB at 135 kB/s)
[INFO] --- compiler:3.13.0:compile (default-compile) @ smart-task-manager ---
Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-compiler-api/2.15.0/plexus-compiler-api-2.15.0.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-compiler-api/2.15.0/plexus-compiler-api-2.15.0.jar (29 kB at 529 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-compiler-manager/2.15.0/plexus-compiler-manager-2.15.0.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-compiler-manager/2.15.0/plexus-compiler-manager-2.15.0.jar (5.2 kB at 69 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-compiler-javac/2.15.0/plexus-compiler-javac-2.15.0.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-compiler-javac/2.15.0/plexus-compiler-javac-2.15.0.jar (26 kB at 306 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-compiler-javac/2.15.0/plexus-compiler-javac-2.15.0.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-compiler-javac/2.15.0/plexus-compiler-javac-2.15.0.jar (26 kB at 306 kB/s)
[INFO] Recompiling the module because of changed source code.
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 8 source files with javac [debug target 21] to target\classes
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 4.929 s
[INFO] Finished at: 2026-06-21T20:17:01+02:00
[INFO]
PS C:\Users\zaid\Desktop\Smart Task Manager>

```

```

Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-compiler-maven
Downloaded from central: https://repo.maven.apache.org/maven2/com/thoughtworks/qdox/qdox/2.0.3/qdox-2.
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-compiler-java
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-utils/4.0.0/p
[INFO] Recompiling the module because of changed source code.
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform depend
[INFO] Compiling 8 source files with javac [debug target 21] to target\classes
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 4.929 s
[INFO] Finished at: 2026-06-21T20:17:01+02:00
[INFO] -----
PS C:\Users\zaid\Desktop\Smart Task Manager> java -cp target/classes Main
=====
SMART TASK MANAGER
=====

----- MAIN MENU -----
1 - Create a task
2 - Show all tasks
3 - Delete a task
4 - Mark a task as completed
5 - Sort tasks by priority
6 - Sort tasks by due date
0 - Exit
-----
Choose an option: |

```

```

at Main.main(Main.java:5)
PS C:\Users\zaid\Desktop\Smart Task Manager> java -cp target/classes TaskManagerManualTest
=== TaskManager Manual Tests ===
PASS: Add task
PASS: Delete task with existing ID
PASS: Delete task with non-existing ID
PASS: Complete task
PASS: Sort tasks by priority
PASS: Sort tasks by due date

Passed: 6
Failed: 0
PS C:\Users\zaid\Desktop\Smart Task Manager> |

```

Welche Ausgaben sind erfolgt?

Maven wurde zuerst in PowerShell nicht erkannt. Nach dem Setzen von `MAVEN_HOME` und `PATH` konnte Maven erfolgreich gestartet werden. Danach wurde das Projekt mit `mvn compile` kompiliert. Am Ende erschien `BUILD SUCCESS`.

Welche Erfahrung habe ich gemacht?

Ich habe gelernt, dass Maven nur funktioniert, wenn es richtig im Systempfad eingetragen ist. Außerdem muss die `pom.xml` im richtigen Projektordner liegen. Nach der Korrektur konnte Maven das Projekt erfolgreich bauen.

2. CI/CD mit GitHub Actions

Warum habe ich dieses Werkzeug gewählt?

GitHub Actions wurde gewählt, weil es direkt in GitHub integriert ist und automatisch Builds nach einem Push ausführen kann. Dadurch kann geprüft werden, ob das Projekt auch außerhalb meines eigenen Rechners gebaut werden kann.

Welche Befehle habe ich ausgeführt?

- Erstellung der Datei `.github/workflows/java-ci.yml`
- `git status`
- `git add .`
- `git commit -m "add GitHub Actions workflow"`
- `git push origin main`

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
PS C:\Users\zaid\Desktop\Smart Task Manager> mkdir .github

Directory: C:\Users\zaid\Desktop\Smart Task Manager

Mode                LastWriteTime         Length Name
----                -
d-----          6/21/2026   9:44 PM             .github

PS C:\Users\zaid\Desktop\Smart Task Manager> mkdir .github\workflows

Directory: C:\Users\zaid\Desktop\Smart Task Manager\.github

Mode                LastWriteTime         Length Name
----                -
d-----          6/21/2026   9:44 PM             workflows

PS C:\Users\zaid\Desktop\Smart Task Manager> New-Item -ItemType File -Path ".github\workflows\java-ci.yml"

Directory: C:\Users\zaid\Desktop\Smart Task Manager\.github\workflows

Mode                LastWriteTime         Length Name
----                -
-a-----          6/21/2026   9:46 PM             0 java-ci.yml

PS C:\Users\zaid\Desktop\Smart Task Manager> cd "C:\Users\zaid\Desktop\Smart Task Manager"
PS C:\Users\zaid\Desktop\Smart Task Manager>
PS C:\Users\zaid\Desktop\Smart Task Manager> New-Item -ItemType Directory -Force -Path ".github\workflows"

Directory: C:\Users\zaid\Desktop\Smart Task Manager\.github

Mode                LastWriteTime         Length Name
----                -
d-----          6/21/2026   9:46 PM             workflows
```

```
PS C:\Users\zaid\Desktop\Smart Task Manager>
PS C:\Users\zaid\Desktop\Smart Task Manager> New-Item -ItemType File -Force -Path ".github\workflows\java-ci.yml"

Directory: C:\Users\zaid\Desktop\Smart Task Manager\.github\workflows

Mode                LastWriteTime         Length Name
----                -
-a-----          6/21/2026   9:46 PM             0 java-ci.yml
```

```
! java-ci.yml X
.github > workflows > ! java-ci.yml
3   on:
6     pull_request:
8
9   jobs:
10    build:
11      runs-on: ubuntu-latest
12
13      steps:
14        - name: Repository auschecken
15          uses: actions/checkout@v5
16
17        - name: Java 21 einrichten
18          uses: actions/setup-java@v5
19          with:
20            distribution: 'temurin'
21            java-version: '21'
22
23        - name: Maven Build ausführen
24          run: mvn compile
25
26        - name: Manuelle Tests ausführen
27          run: java -cp target/classes TaskManagerManualTest

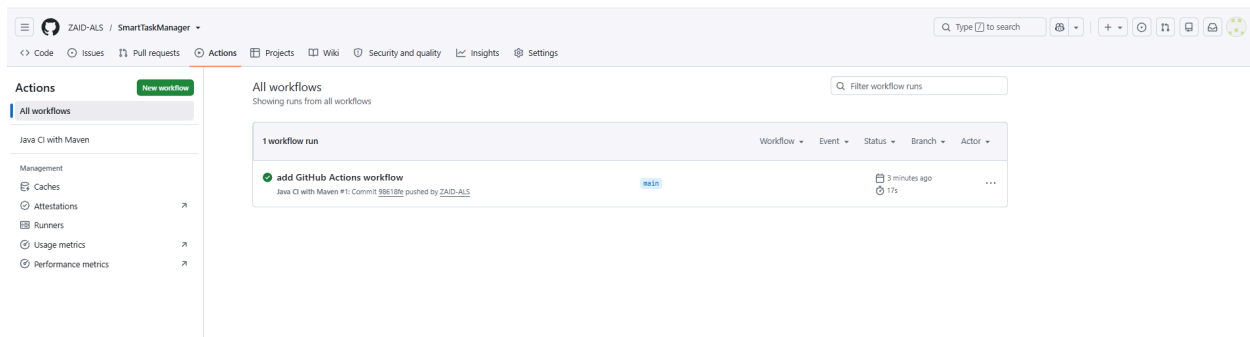
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

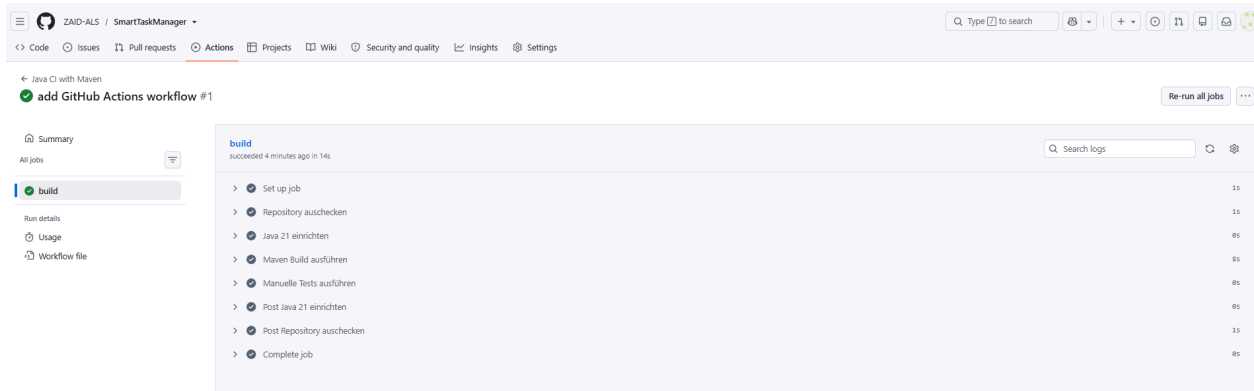
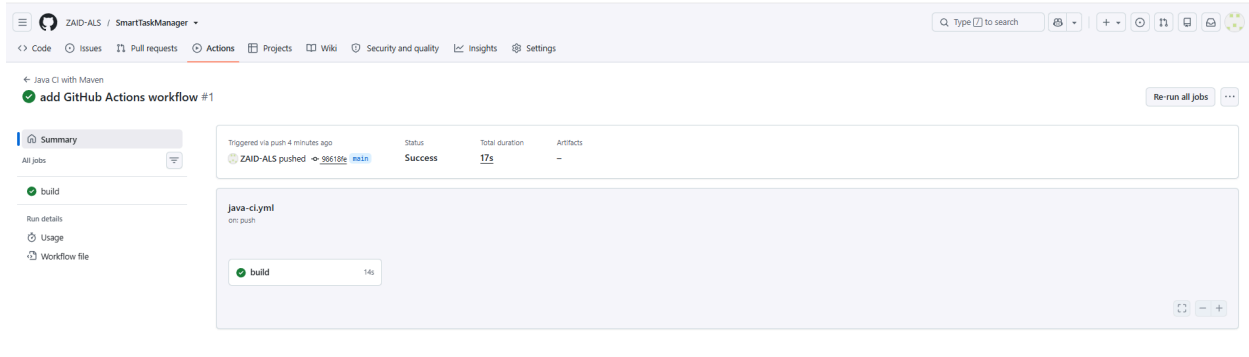
PS C:\Users\zaid\Desktop\Smart Task Manager> git status
On branch main
Your branch is up to date with 'origin/main'.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
  .github/

nothing added to commit but untracked files present (use "git add" to track)
PS C:\Users\zaid\Desktop\Smart Task Manager> git add .github\workflows\java-ci.yml
PS C:\Users\zaid\Desktop\Smart Task Manager> git commit -m "add GitHub Actions workflow"
[main 98618fe] add GitHub Actions workflow
1 file changed, 27 insertions(+)
create mode 100644 .github/workflows/java-ci.yml
PS C:\Users\zaid\Desktop\Smart Task Manager> git push origin main
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 12 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (5/5), 660 bytes | 660.00 KiB/s, done.
Total 5 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/ZAID-ALS/SmartTaskManager.git
   adc8913..98618fe  main -> main
PS C:\Users\zaid\Desktop\Smart Task Manager> 
```

🕒 Zaid Alsalamah (3 minutes ago)





Welche Ausgaben sind erfolgt?

Nach dem Push wurde der Workflow auf GitHub automatisch gestartet. GitHub Actions hat das Repository ausgecheckt, Java 21 eingerichtet, Maven ausgeführt und die manuelle Testklasse gestartet.

Welche Erfahrung habe ich gemacht?

Ich habe gelernt, dass CI/CD hilft, Fehler früh zu erkennen. Der Build läuft nicht nur lokal auf meinem Rechner, sondern auch automatisch auf GitHub. Dadurch wird der Projektstand besser überprüfbar.